Quantum Computation and Information

Lecture 3: Quantum Algorithms

- Deutsch's algorithm
- Grover's algorithm
- quantum Fourier transformation
- period finding
- phase estimation
- quantum factoring

Deutsch's algorithm

- a blackbox device that computes a function $f : \{0,1\} \to \{0,1\}$
  - four possible functions

	f_0	f_1	f_2	f_3
0	0	0	1	1
1	0	1	0	1

  - determine if f is a constant function, or a balanced function (each possible function value appears the same number of times)

- two computations are required classically

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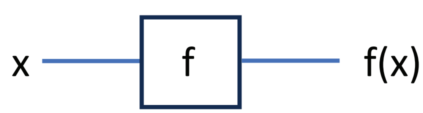
  - one-qubit gate?



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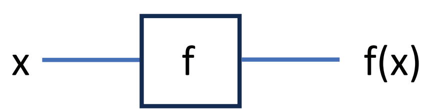

no, e.g., $f_0 = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ not unitary

Deutsch's algorithm

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  determine if f is a constant function, or a balanced function (each possible function value appears the same number of times)

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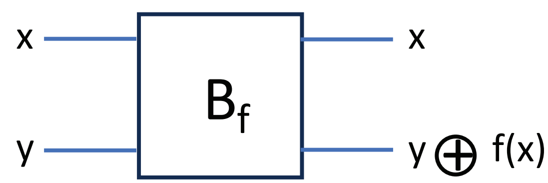
- what if in the setting of quantum computing?

  – two-qubit gate

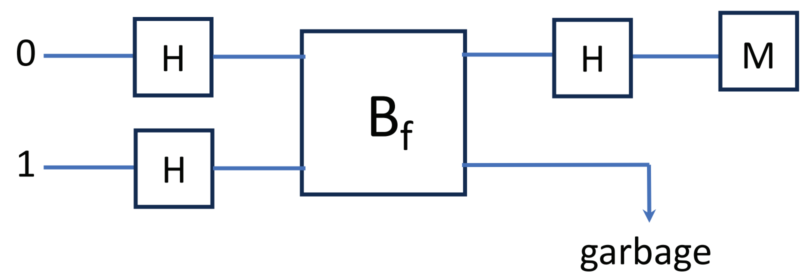


$$\text{e.g., } f_2 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \text{ unitary}$$

Deutsch's algorithm

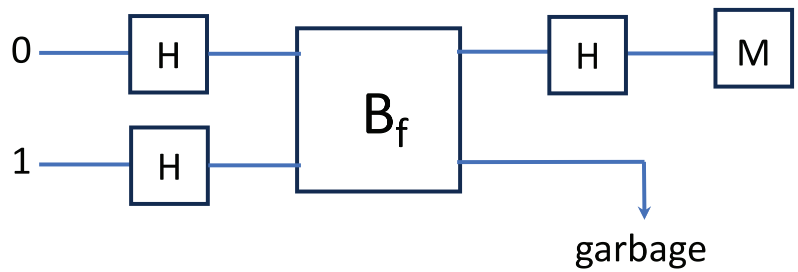
- a blackbox transformation B_f for some function $f : \{0,1\} \to \{0,1\}$ determine if f is a constant function, or a balanced function

- only one application of B_f is enough (v.s. two in classical computing)



  - state of the first qubit before measurement $(-1)^{f(0)} |f(0) \oplus f(1)\rangle$
  - measurement outcome is 0 if constant function and 1 balanced function
  - quantum advantage due to quantum parallel: input is a superposition of $|0\rangle$ and $|1\rangle$ and output is sensitive to both $f(0)$ and $f(1)$

Deutsch's algorithm



$$|0\rangle\,|1\rangle \Longrightarrow [\tfrac{1}{\sqrt{2}}(|0\rangle + |1\rangle)][\tfrac{1}{\sqrt{2}}(|0\rangle - |1\rangle)] = \tfrac{1}{2}\,|0\rangle\,(|0\rangle - |1\rangle) + \tfrac{1}{2}\,|1\rangle\,(|0\rangle - |1\rangle)$$

$$\Longrightarrow \tfrac{1}{2}\,|0\rangle\,(|0 \oplus f(0)\rangle - |1 \oplus f(0)\rangle) + \tfrac{1}{2}\,|1\rangle\,(|0 \oplus f(1)\rangle - |1 \oplus f(1)\rangle)$$

$$= \tfrac{1}{2}(-1)^{f(0)}\,|0\rangle\,(|0\rangle - |1\rangle) + \tfrac{1}{2}(-1)^{f(1)}\,|1\rangle\,(|0\rangle - |1\rangle)$$

$$= (\tfrac{1}{\sqrt{2}}(-1)^{f(0)}\,|0\rangle + \tfrac{1}{\sqrt{2}}(-1)^{f(1)}\,|1\rangle)(\tfrac{1}{\sqrt{2}}\,|0\rangle - \tfrac{1}{\sqrt{2}}\,|1\rangle)$$

$$\tfrac{1}{\sqrt{2}}(-1)^{f(0)}\,|0\rangle + \tfrac{1}{\sqrt{2}}(-1)^{f(1)}\,|1\rangle = \tfrac{1}{\sqrt{2}}(-1)^{f(0)}(|0\rangle + (-1)^{f(0)\oplus f(1)}\,|1\rangle$$

$$\Longrightarrow \tfrac{1}{\sqrt{2}}(-1)^{f(0)}\,|f(0) \oplus f(1)\rangle$$

$\Longrightarrow$measurement: $f(0) \oplus f(1)$, which is 0 if $f(0) = f(1)$ and 1 otherwise

Grover's algorithm

unstructured search

given a function $f : \{0,1\}^n \to \{0,1\}$ with

$$f(x) = \{ \begin{array}{ll} 0 & x \neq w \\ 1 & x = w \end{array}$$

for some unknown w, find out w

Grover's algorithm

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$$f(x) = \{ \begin{array}{ll} 0 & x \neq w \\ 1 & x = w \end{array}$$

for some unknown w, find out w

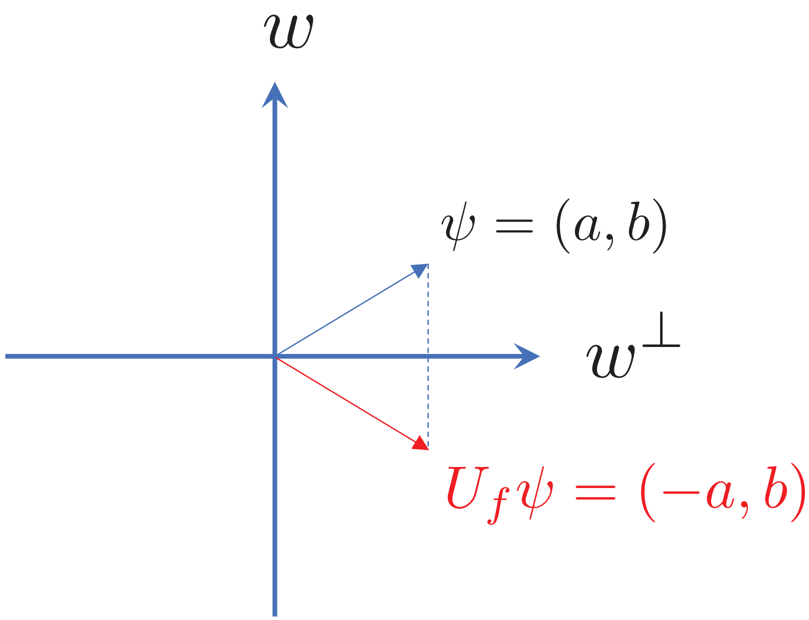
- classically
  - 2^n queries required at the worse case
  - $\Omega((1-\epsilon)2^n)$ queries required for randomized algorithm with probability at least $1-\epsilon$
- quantumly, $O(\sqrt{2^n})$ queries using Grover's algorithm (a quadratic speedup)

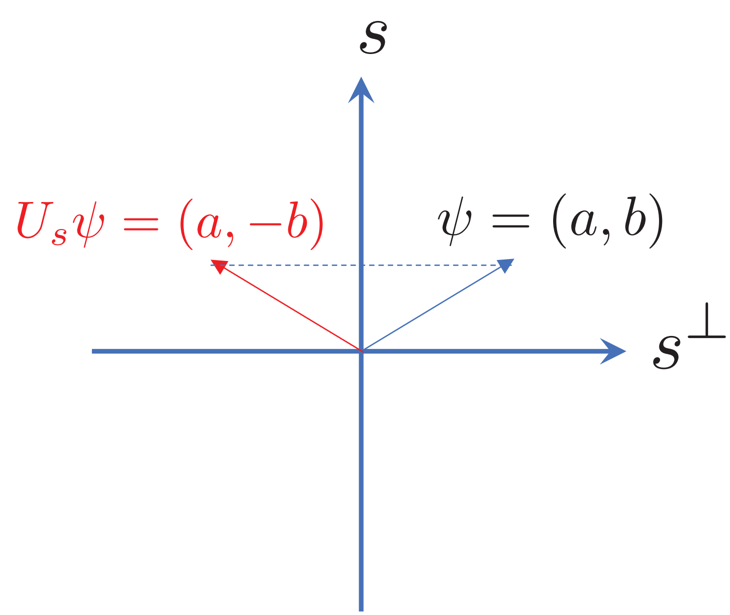
Grover's algorithm

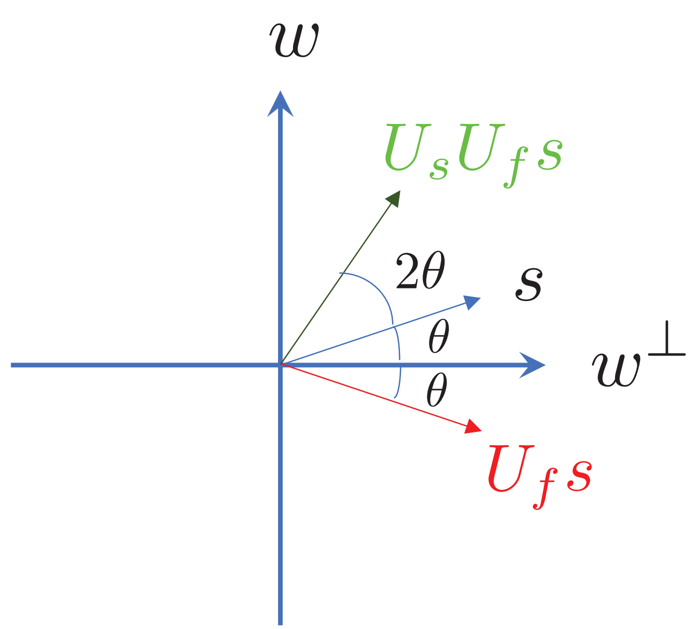
- function $f : \{0,1\}^n \to \{0,1\}$ implemented by a unitary transformation U_f
$$U_f \left|x\right\rangle \left|a\right\rangle = \left|x\right\rangle \left|a \oplus f(x)\right\rangle$$
where $x \in \{0,1\}^n$ and $a \in \{0,1\}$
  - for function $f : \{0,1\}^n \to \{0,1\}^m$, $U_f \left|x\right\rangle \left|a\right\rangle = \left|x\right\rangle \left|a \oplus f(x)\right\rangle$ with $x \in \{0,1\}^n$, $a \in \{0,1\}^m$ and $\oplus$ bitwise XOR

- U_f turned into a 'phase oracle': $U_f \left|x\right\rangle \left|-\right\rangle = (-1)^{f(x)} \left|x\right\rangle \left|-\right\rangle$, where $(-1)^{f(x)}$ equals 1 if $x \neq w$ and -1 if $x = w$

- if ignoring the output register, $U_f = I - 2\left|w\right\rangle \left\langle w\right|$ on the input
  - given n-qubit state $\left|\psi\right\rangle = a\left|w\right\rangle + b\left|w^\perp\right\rangle$, $U_f \left|\psi\right\rangle = -a\left|w\right\rangle + b\left|w^\perp\right\rangle$
  - a reflection on 'plane' $\left|w^\perp\right\rangle$


w
$\psi = (a, b)$
$w^{\perp}$
$U_f\psi = (-a, b)$


s
$U_s\psi = (a, -b)$
$\psi = (a, b)$
$s^{\perp}$


w
$U_sU_f s$
2θ
s
θ
$w^{\perp}$
θ
$U_f s$

Grover's algorithm

Grover's algorithm

- prepare uniform superposition $|s\rangle = H^{\otimes n}|00\ldots0\rangle = \frac{1}{\sqrt{2^n}}\sum_{x\in\{0,1\}^n}|x\rangle$

- apply Grover iteration in succession: $U_G = U_s U_f$ with $U_s = 2|s\rangle\langle s| - I$
  - U_f is the query, U_s a reflection about the 'axis' $|s\rangle$
  - $U_s = H^{\otimes n}(2|00\ldots0\rangle\langle 00\ldots0| - I)H^{\otimes n}$ can be realized with a circuit of size $O(n)$
  - denote by θ the angle between $|s\rangle$ and $|w^\perp\rangle$, each U_G rotates the state vector counterclockwise by 2θ
    * $\sin\theta = \langle s|w\rangle = 1/\sqrt{2^n}$
    * each iteration enhances in the direction of $|w\rangle$

- the required number T of Grover iterations?

Grover's algorithm

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- the required number of Grover iterations $T = O(\sqrt{2^n})$
  - $\theta \approx 1/\sqrt{2^n}$
  - $(1+2T)\theta = \pi/2 + \delta$ with $|\delta| \leq \theta$
  - probability to find w: $P(w) = \cos^2\delta \geq 1 - \delta^2 \geq 1 - \frac{1}{2^n}$

Grover's algorithm

- what if there are $r > 1$ values of x such that $f(x) = 1$?

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  - define $|\varphi\rangle = \frac{1}{\sqrt{r}} \sum_i |w_i\rangle$, $\sin\theta = \langle s|\varphi\rangle = \sqrt{r/2^n}$
  - $T = O(\sqrt{\frac{2^n}{r}})$

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  - $T = O(\sqrt{\frac{2^n}{r}})$

- what if no prior knowledge of r?
  - the success probability oscillates in T with period $\sim \frac{\pi}{2}\sqrt{\frac{2^n}{r}}$
  - choose number of queries randomly between 0 and $\sim \frac{\pi}{4}\sqrt{2^n}$, one is found with probability greater than 1/2 if one exists.
  - $T = O(\sqrt{2^n})$ with high probability

quantum Fourier transformation

$$\sum_x f(x)\,|x\rangle \to \sum_y \left(\frac{1}{\sqrt{M}} \sum_x e^{i2\pi xy/M} f(x)\right)|y\rangle$$

- $x, y \in \{0, 1, \cdots, M-1\}$
- classical discrete Fourier transform applied to vector of probability amplitude of quantum state
  - classically, of complexity $O(M^2)$ by definition and $O(M \log M)$ by FFT
- a unitary transformation!
  - quantumly, of complexity $(\log M)^2$ (an exponential gap, with the help of quantum parallel!)

quantum Fourier transformation

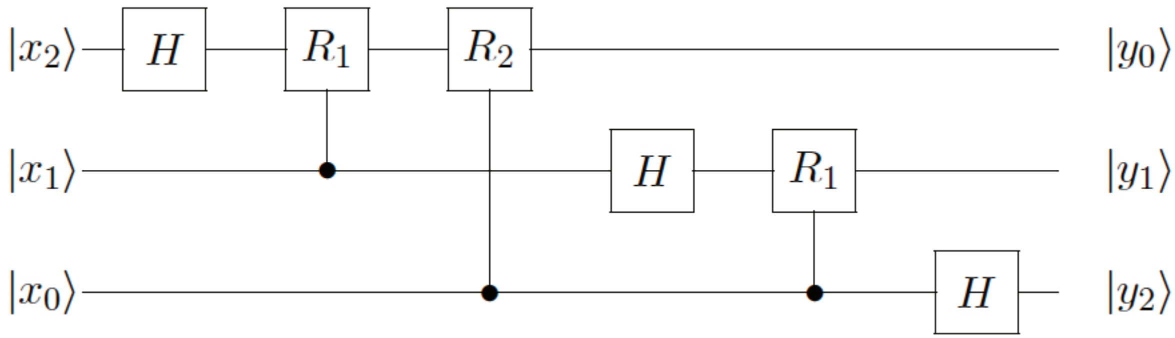
$$QFT\,|x\rangle = \frac{1}{\sqrt{2^m}} \sum_{y=1}^{2^m-1} e^{i\frac{2\pi xy}{2^m}}\,|y\rangle$$

- $x, y \in \{0, 1, \cdots, 2^m - 1\}$, index for m-qubit state
- let $x \equiv x_{m-1}x_{m-2}\ldots x_0$ and $y \equiv y_{m-1}y_{m-2}\ldots y_0$
- $\frac{xy}{2^m} \equiv y_{m-1}(.x_0) + y_{m-2}(.x_1x_0) + \cdots + y_0(.x_{m-1}x_{m-2}\cdots x_0)\ (mod\ 1)$
- QFT takes each computational basis to a separable state of m qubit

$$\begin{aligned} QFT\,|x\rangle &= \tfrac{1}{\sqrt{2^m}} \textstyle\sum_{y_{m-1}} \sum_{y_{m-2}} \cdots \sum_{y_0} e^{i\frac{2\pi xy}{2^m}}\,|y_{m-1}y_{m-2}\cdots y_0\rangle \\ &= \tfrac{1}{\sqrt{2^m}}(|0\rangle + e^{i2\pi(.x_0)}\,|1\rangle)(|0\rangle + e^{i2\pi(.x_1x_0)}\,|1\rangle) \\ &\quad \cdots (|0\rangle + e^{i2\pi(.x_{m-1}x_{m-2}\cdots x_0)}\,|1\rangle) \end{aligned}$$

quantum Fourier transformation

$$
\begin{aligned}
QFT\,|x\rangle &= \frac{1}{\sqrt{2^m}} \sum_{y=0}^{2^m-1} e^{i\frac{2\pi xy}{2^m}}\,|y\rangle \\
&= \frac{1}{\sqrt{2^m}}(|0\rangle + e^{i2\pi(.x_0)}\,|1\rangle)(|0\rangle + e^{i2\pi(.x_1x_0)}\,|1\rangle) \\
&\quad \cdots (|0\rangle + e^{i2\pi(.x_{m-1}x_{m-2}\cdots x_0)}\,|1\rangle)
\end{aligned}
$$



- $H : |x_k\rangle \to \frac{1}{\sqrt{2}}(|0\rangle + e^{i2\pi(.x_k)}\,|1\rangle)$ and $R_d = \begin{pmatrix} 1 & 0 \\ 0 & e^{\frac{i\pi}{2^d}} \end{pmatrix}$
- m H-gates and $m(m-1)/2$ controlled R's, so QFT circuit is of size $m^2 = (\log M)^2$
- complexity can be brought down to $\log M$ if measuring right after QFT

period finding

given function $f : \{0,1\}^n \to \{0,1\}^m$ that has an unknown period r (with $1 \ll r < 2^m \ll 2^n$)

$$f(x) = f(x + kr)$$

where $k \in \mathbb{Z}_+$ (the set of nonnegative integers), find out r

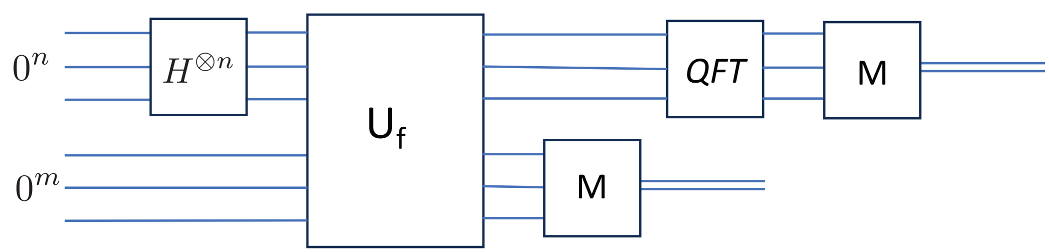
- classically, a hard problem; e.g., if $r = O(2^{n/2})$, then $O(2^{n/4})$ queries required before likely finding two x values with the same function value

- quantumly, r can be found in $poly(n)$ time by applying quantum Fourier transformation (QFT)

period finding

- function $f : \{0,1\}^n \to \{0,1\}^m$ implemented by a unitary transformation U_f
$$U_f |x\rangle |a\rangle = |x\rangle |a \oplus f(x)\rangle$$
where $x \in \{0,1\}^n$, $a \in \{0,1\}^m$ and $\oplus$ bitwise XOR

- algorithm



- before measurement

$$|0^n\rangle |0^m\rangle \Longrightarrow \left(\frac{1}{\sqrt{2^n}} \sum_{x\in\{0,1\}^n} |x\rangle\right) |0^m\rangle \Longrightarrow \frac{1}{\sqrt{2^n}} \sum_{x\in\{0,1\}^n} |x\rangle |f(x)\rangle$$

- supposed the measurement outcome on output register is $|f(x_0)\rangle$ for some $0 \leq x_0 < r$, then it prepares the input register the state

$$\frac{1}{\sqrt{L}} \sum_{j=0}^{L-1} |x_0 + jr\rangle$$

  where L is the number of x values with function value $f(x_0)$ and $2^n - r \leq x_0 + (L-1)r < 2^n$

- apply QFT

$$\frac{1}{\sqrt{2^n L}} \sum_{y=0}^{2^n-1} e^{i2\pi x_0 y/2^n} \sum_{j=0}^{L-1} e^{i2\pi jry/2^n} |y\rangle$$

  – it is hard to find a needle, but it is easy to find periodic needles

period finding

- probability of obtaining outcome y

$$Prob(y) = \frac{L}{2^n} \left| \frac{1}{L} \sum_{j=0}^{L-1} e^{i2\pi jry/2^n} \right|^2$$

  - independent of x_0; distribution strongly peaked at values of y with $yr/2^n$ close to an integer
  - if $2^n/r$ happens to be an integer, then $2^n/r = L$ and

$$Prob(y) = \frac{1}{r} \left| \frac{1}{L} \sum_{j=0}^{L-1} e^{i2\pi jy/L} \right|^2 = \begin{cases} \frac{1}{r} & y \text{ is multiple of } L \\ 0 & \text{otherwise} \end{cases}$$

    * constructive interference if y is multiple of $2^n/r$ and destructive interference otherwise
    * sample uniformly y such that y is multiple of $2^n/r$

period finding

- probability of obtaining outcome y

$$Prob(y) = \frac{L}{2^n} | \frac{1}{L} \sum_{j=0}^{L-1} e^{i2\pi jry/2^n} |^2$$

  - distribution strongly peaked at values of y with $yr/2^n$ close to an integer
  - in general, let $\theta_y = \frac{yr(mod\ 2^n)}{2^n} 2\pi$

$$Prob(y) = \frac{L}{2^n} | \frac{1}{L} \sum_{j=0}^{L-1} e^{i\theta_y j} |^2 = \frac{1}{2^n L} | \frac{e^{iL\theta_y} - 1}{e^{i\theta_y} - 1} |^2$$

    * r values of y in $\{0, 1, \cdots, 2^n - 1\}$ satisfying $-\frac{r}{2} \leq yr\ (mod\ 2^n) \leq \frac{r}{2}$ and thus $-\frac{r}{2^n}\pi \leq \theta_y \leq \frac{r}{2^n}\pi$
    * note that $L - 1 < \frac{2^n}{r}$, so all terms in sum for $Prob(y)$ are in the same half plane and interfere constructively

period finding

- probability of obtaining outcome y

$$Prob(y) == \frac{1}{2^n L}|\frac{e^{iL\theta_y}-1}{e^{i\theta_y}-1}|^2 = \frac{1}{2^n L}\frac{\sin^2(L\theta_y/2)}{\sin^2(\theta_y/2)} \geq \frac{1}{r}\frac{4}{\pi^2}$$

  - r values of y satisfying $-\frac{r}{2} \leq yr \ (mod \ 2^n) \leq \frac{r}{2}$, so probability for sampling such y is greater than $4/\pi^2 \approx 0.405$
    * equivalently, $|\frac{y}{2^n} - \frac{k}{r}| \leq \frac{1}{2}\frac{1}{2^n}$, $k \in \{0, 1, \cdots, r-1\}$
  - given that $r < 2^m$, if $2^n \geq (2^m)^2$, a unique k/r can be inferred
    * smallest space between rational numbers with denominator smaller than 2^m is $1/(2^m)^2$
    * can be efficiently extracted by the continued fraction method
  - r or its factor can be obtained from k/r

phase estimation

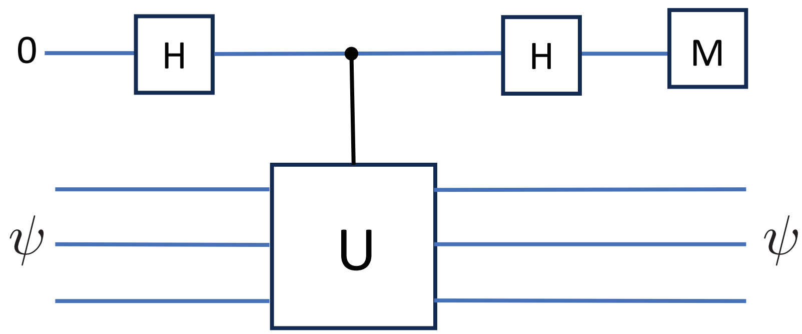
- let $N = 2^n$, an unitary transformation $U(N)$ has an orthonormal basis of eigenvectors $|\psi_1\rangle, |\psi_2\rangle, \cdots, |\psi_N\rangle$, with associated eigenvalues $e^{2\pi i\theta_1}, e^{2\pi i\theta_2}, \cdots, e^{2\pi i\theta_N}$

- **phase estimation:** given an unitary transformation U and one of its eigenvector $|\psi\rangle$, find out the corresponding eigenvalue, i.e., θ such that

$$U|\psi\rangle = e^{2\pi i\theta}|\psi\rangle$$

phase estimation

- first try



$$|0\rangle \Longrightarrow \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \Longrightarrow \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i\theta}\,|1\rangle)$$

$$\Longrightarrow \tfrac{1}{2}(1 + e^{2\pi i\theta})\,|0\rangle + \tfrac{1}{2}(1 - e^{2\pi i\theta})\,|1\rangle$$

  – measurement outcome of control bit

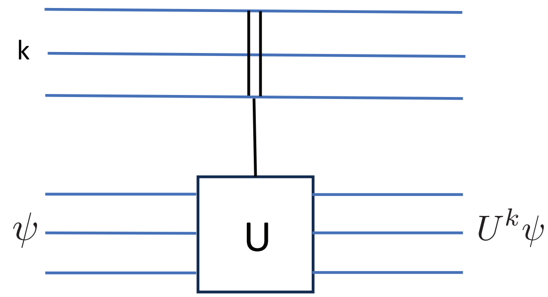
$$\text{Prob}(0) = |\tfrac{1}{2}(1 + e^{2\pi i\theta})|^2 = \tfrac{1+\cos 2\pi\theta}{2}$$

$$\text{Prob}(1) = |\tfrac{1}{2}(1 - e^{2\pi i\theta})|^2 = \tfrac{1-\cos 2\pi\theta}{2}$$

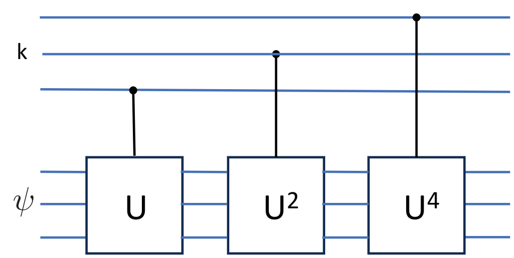
  – $O(2^m)$ measurements required to estimate $\cos 2\pi\theta$ within m bits of accuracy

phase estimation

- circuit $\Lambda_m(U) : |k\rangle\,|\psi\rangle \to |k\rangle\,U^k\,|\psi\rangle$



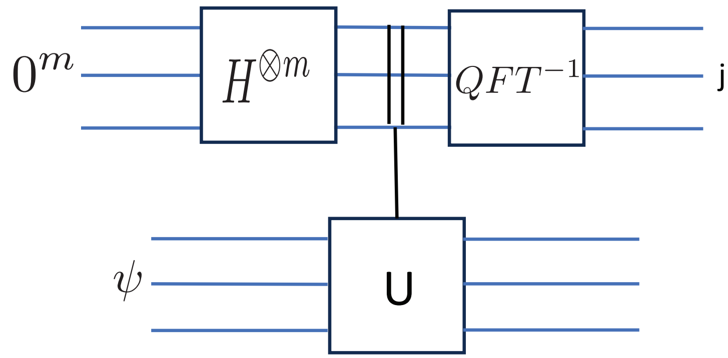
  - $k \in \{0, 1, \cdots, M-1\}$ with $M = 2^m$; m-qubit states indexed by k
  - implement using control-U, control-U^2, ... , control-$U^{2^{m-1}}$ gates



- estimating θ within m-bit of accuracy is to estimate integer $j < M$ such that $\frac{j}{M}$ is the closest approximation to θ

phase estimation

- efficient estimation



$$|0^m\rangle\,|\psi\rangle \Longrightarrow \left(\frac{1}{\sqrt{M}}\sum_{k=0}^{M-1}|k\rangle\right)|\psi\rangle \Longrightarrow \left(\frac{1}{\sqrt{M}}\sum_{k=0}^{M-1}e^{i2\pi\theta k}\,|k\rangle\right)|\psi\rangle$$

$$\Longrightarrow \sum_{j=0}^{M-1}\left(\frac{1}{M}\sum_{k=0}^{M-1}e^{i2\pi k(\theta-\frac{j}{M})}\right)|j\rangle$$

– $Prob(j) = 1$ if $\theta = \frac{j}{M}$
– $Prob(j) \geq 4/\pi^2$ if $\frac{j}{M}$ is the closest approximation.

factoring

given positive integer N, find a prime factorization $N = p_1^{k_1} p_2^{k_2} \cdots p_m^{k_m}$, where p_i's are prime numbers and k_i's positive integers

- believed to be a hard problem classically (NP but not NP-complete), no known $ploy(\log N)$ algorithm

- its hardness is the basis for the security of the widely used RSA (Rivest, Shamir, and Adleman) scheme for public key cryptography

- Shor's algorithm has a $O((\log N)^3)$ complexity

- order finding: quantum part of Shor's algorithm is a subroutine for finding period of an efficiently computable function

factoring

order finding

- multiplicative group mod N: $\mathbb{Z}_N^* = \{a \in \mathbb{Z} | 1 \leq a < N, GCD(N, a) = 1\}$
  - a set S with an operation '$\cdot$' satisfying
    - identity element: there exists an element e such that $a \cdot e = e \cdot a = a, \ \forall a \in S$
    - inverse: for any $a \in S$ there exists $b \in S$ such that $a \cdot b = b \cdot a = e$
    - associativity: for $a, b, c \in S$, $(a \cdot b) \cdot c = a \cdot (b \cdot c)$
  - for $a \in \mathbb{Z}_N^*$, define $S_a = \{ab \ (mod \ N) | b \in \mathbb{Z}_N^*\}$
    - S_a has distinct elements
    - there exists a b such that $ab = 1 \ (mod \ N)$
    - there exists an integer $r \in \mathbb{Z}_{++}$ such that $a^r = 1 \ (mod \ N)$

- for a number $a < N$ coprime to N, its order is defined as the smallest positive integer r such that $a^r = 1 \ (mod \ N)$

- the order of a is the period of modular exponential function $f_{N,a}(x) = a^x \ (mod \ N)$

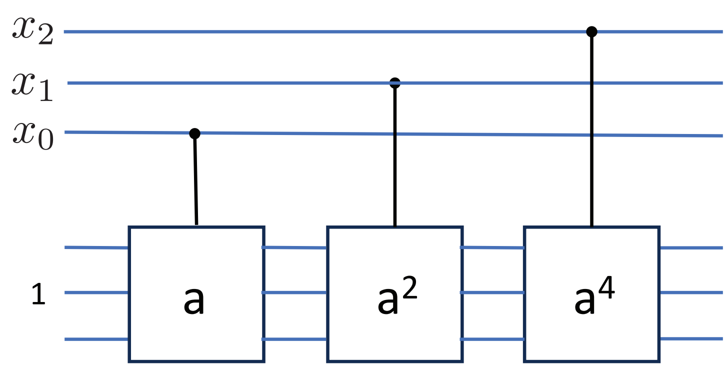
- period finding algorithm used to find the order r if modular exponential can be efficiently computed

- modular exponential can be computed in $O((\log N)^3)$ time by repeated squaring

  – let $x \equiv x_{m-1}x_{m-2}\ldots x_0$

$$a^x \ (mod \ N) = (a^{2^{m-1}})^{x_{m-1}}(a^{2^{m-2}})^{x_{m-2}}\cdots(a)^{x_0} \ (mod \ N)$$

  – compute a^{2^k} by squaring k times: $a \rightarrow a^2 \rightarrow a^4 \rightarrow \cdots \rightarrow a^{2^k}$



factoring

Shor's algorithm

- randomly choose $a < N$
- if the greatest common divisor $GCD(a, N) \neq 1$, then $GCD(a, N)$ is nontrivial factor of N (done!)
  - GCD computed by Euclidean algorithm in $O((\log N)^3)$ time
- if $GCD(a, N) = 1$, find the order r of a via quantum period finding algorithm
- if r is even, N divides $(a^{r/2} + 1)(a^{r/2} - 1)$ and
  - either N divides $a^{r/2} + 1$
  - or, factors of N split between $a^{r/2} + 1$ and $a^{r/2} - 1$ and then $GCD(a^{r/2} \pm 1, N)$ are nontrivial factors of N (done!)

- an example: $N = 15$ and $a = 7$
  * $a\ (mod\ N) = 7$, $a^2\ (mod\ N) = 4$, $a^3\ (mod\ N) = 13$ and $a^4\ (mod\ N) = 1$, so a has an order of $r = 4$
  * $a^{r/2} - 1 = 48$ and $a^{r/2} + 1 = 50$, and $GCD(a^{r/2} - 1, N) = 3$ and $GCD(a^{r/2} + 1, N) = 5$

- the success probability is at least $1/2$

- apply the above recusively

factoring

factoring via phase estimation (by Kitaev)

- for $a < N$ coprime to N, consider unitary transformation U_a

$$U_a \left| x \right\rangle = \left| ax \ (mod \ N) \right\rangle$$

  – the order r of a implies $U_a^r = I$ and thus eigenvalues of U_a are

$$e^{\frac{i2\pi k}{r}}, \ k \in \{0, 1, \cdots, r-1\}$$

- phase estimation determines a value of k/r, the same as in period finding